CIE v2.3.2

HARDENING ADDENDUM

6 Resilience, Governance, and AI-Readiness Patches

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Patches CIE v2.3.1 Enforcement Dev Spec + Developer Build Pack

Patch Summary: v2.3.1 → v2.3.2

Cross-engine review (Gemini + Perplexity) identified 6 production-critical gaps. This addendum patches each one with implementable specs that integrate directly into the existing DB schema, OpenAPI endpoints, and operational workflows.

#	Patch	Risk if Missing	What It Adds	Integrates With
1	Fail-Soft Vector Validation	**CRITICAL**	VECTOR_PENDING status, retry cron, degraded-save mode	DB: sku_gate_status, OpenAPI: /validate, N8N
2	AI Audit Degradation Rules	**HIGH**	Engine quorum, partial-run handling, fallback scoring	DB: ai_audit_runs, OpenAPI: /audit/run, N8N
3	Decomposed AI Readiness Scoring	**MEDIUM**	Per-intent readiness components, weighted formula	DB: channel_readiness, Dashboard
4	Mandatory FAQ Templates per Product Class	**MEDIUM**	Standard question sets, readiness gate, citability boost	DB: sku_faq_responses, CMS UI, JSON-LD
5	Cluster Governance Playbook	**LOW-MED**	Change proposal workflow, impact check, quarterly cadence	RBAC, audit_log, Operational SOP
6	Tier-Mode UX Copy & Banners	**LOW-MED**	CMS banner text per tier, tooltip copy, editor guidance	PHP CMS frontend, training materials

Confidence: v2.3.1 = 93-95% → v2.3.2 = 97-98%. Remaining 2-3% = multi-language stubs + brand knowledge graph (deferred to v3.0).

Patch 1: Fail-Soft Vector Validation

PROBLEM: v2.3.1 specifies cosine similarity ≥0.72 on every save/publish, but never defines what happens when the embedding API (OpenAI text-embedding-3-small) is down. Two bad options exist: fail closed (120 staff locked out) or fail open (bypass enforcement). Neither is acceptable.

1.1 The Fail-Soft Protocol

A middle path: allow save with degraded status, block publish, auto-retry asynchronously.

Scenario	Save?	Publish?	VECTOR Gate Status	System Behaviour
Embedding API OK	**YES**	**YES***	pass / fail (normal)	Normal validation. Score stored. Gate enforced.
API timeout (>2s)	**YES**	**BLOCKED**	pending	Save allowed. VECTOR gate = pending. Publish blocked. Queued for retry.
API 500/503 error	**YES**	**BLOCKED**	pending	Same as timeout. Logged as embedding_api_error in audit_log.
API down >1 hour	**YES**	**BLOCKED**	pending	Admin alerted. All pending SKUs queued. Dashboard shows 'Embedding Service Degraded'.
Retry succeeds	N/A	**UNBLOCKED**	pass / fail (resolved)	Gate updated to pass/fail. If pass, publish enabled. If fail, editor notified.

***** Publish only allowed when ALL gates show 'pass' or 'not_applicable'. 'pending' blocks publish.

1.2 Database Changes

```
-- Add 'pending' to gate status enum
ALTER TABLE sku_gate_status
  ALTER COLUMN status TYPE VARCHAR(20);
-- New valid values: pass, fail, not_applicable, pending

-- New table for retry queue
CREATE TABLE vector_retry_queue (
  id            UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  sku_id        VARCHAR(50) NOT NULL REFERENCES sku_master(sku_id),
  description   TEXT NOT NULL,           -- Text to embed
  cluster_id    VARCHAR(50) NOT NULL,    -- For similarity check
  retry_count   SMALLINT NOT NULL DEFAULT 0,
  max_retries   SMALLINT NOT NULL DEFAULT 5,
  next_retry_at TIMESTAMP NOT NULL DEFAULT (NOW() + INTERVAL '5 minutes'),
  status        ENUM('queued','processing','resolved','failed') DEFAULT 'queued',
  error_message VARCHAR(500),
  created_at    TIMESTAMP NOT NULL DEFAULT NOW(),
  resolved_at   TIMESTAMP
);

CREATE INDEX idx_retry_status ON vector_retry_queue(status, next_retry_at);
```

1.3 Retry Logic (Python Cron — runs every 5 minutes)

```
def process_vector_retry_queue():
    pending = db.query(
        'SELECT * FROM vector_retry_queue'
        ' WHERE status = %s AND next_retry_at <= NOW()'
        ' ORDER BY created_at LIMIT 50',
        ['queued']
    )

    for item in pending:
        try:
            db.update(item.id, status='processing')
            embedding = openai.embed(item.description)  # Attempt API call
            cluster_vec = db.get_cluster_vector(item.cluster_id)
            similarity = cosine_similarity(embedding, cluster_vec)

            gate_status = 'pass' if similarity >= 0.72 else 'fail'
            db.update_gate_status(item.sku_id, 'VECTOR', gate_status,
                                  error_code='CIE_VEC_LOW' if gate_status == 'fail' else
None)
            db.update_sku_content(item.sku_id, vector_similarity=similarity)
            db.update(item.id, status='resolved', resolved_at=now())
            audit_log('vector_retry_resolved', item.sku_id, gate_status)

            if gate_status == 'fail':
                notify_content_owner(item.sku_id, similarity)

        except EmbeddingAPIError as e:
            new_count = item.retry_count + 1
            if new_count >= item.max_retries:
                db.update(item.id, status='failed', error_message=str(e))
                alert_admin(f'Vector retry failed after {new_count} attempts: {item.sku_id}')
            else:
                backoff = min(5 * (2 ** new_count), 60)  # Exponential: 10,20,40,60 min
                db.update(item.id, retry_count=new_count,
                          next_retry_at=now() + timedelta(minutes=backoff),
                          status='queued')
```

1.4 OpenAPI Changes

```
# Add to ValidationResponse.gates.{gate}.status enum:
status:
  type: string
  enum: [pass, fail, not_applicable, pending]  # NEW: pending

# Add to ValidationResponse:
degraded_mode:
  type: boolean
  description: True if any gate returned 'pending' due to service failure
save_allowed:
  type: boolean
  description: True even in degraded mode (save ≠ publish)
publish_allowed:
  type: boolean
  description: False if any gate is 'pending' or 'fail'
```

CRITICAL RULE

A SKU with VECTOR gate = 'pending' CANNOT be published. It CAN be saved. The editor sees: 'Description validation temporarily unavailable. Your changes are saved but publishing is paused until validation completes (typically within 30 minutes).'
This preserves 'no bypass' without causing a total work stoppage.

Patch 2: AI Audit Degradation Rules

PROBLEM: The weekly AI audit queries 4 engines (ChatGPT, Gemini, Perplexity, Google SGE) with 20 questions each. v2.3.1 never defined what happens when 1-2 engines are down, rate-limited, or returning garbage. Without this, decay triggers can fire on incomplete data — destroying trust in the system.

2.1 Engine Quorum Rules

Scenario	Engines OK	Audit Status	Decay Trigger Behaviour
All 4 engines respond	**4/4**	COMPLETE — full confidence	Normal decay logic applies
3 of 4 engines respond	**3/4**	COMPLETE — quorum met	Normal decay logic. Missing engine flagged in notes.
2 of 4 engines respond	**2/4**	PARTIAL — below quorum	Scores recorded. Decay timer PAUSED (not advanced). Admin alerted.
1 or 0 engines respond	**≤1/4**	FAILED — not counted	Run marked as failed. NO scores recorded. Decay timer FROZEN. Reschedule 24hr.

2.2 Per-Engine Failure Handling

```
def run_audit_for_engine(engine, questions):
    results = []
    consecutive_failures = 0

    for q in questions:
        try:
            response = query_engine(engine, q.text, timeout=30)
            score = evaluate_citation(response, q.success_criteria)
            results.append((q.id, score, response[:500]))
            consecutive_failures = 0

        except RateLimitError:
            sleep(exponential_backoff(consecutive_failures))
            consecutive_failures += 1
            if consecutive_failures >= 3:
                # Engine is rate-limiting us — stop, return partial
                return results, 'rate_limited'
            continue  # Retry this question

        except (TimeoutError, APIError) as e:
            consecutive_failures += 1
            if consecutive_failures >= 5:
                return results, 'engine_down'
            results.append((q.id, None, str(e)))  # Null score = skipped

    return results, 'complete'
```

2.3 Partial Score Aggregation

Minimum question coverage: At least 15 of 20 questions must receive a valid score from each responding engine. If <15 scored, that engine is treated as 'failed' for quorum purposes.

Score calculation: Aggregate citation rate computed ONLY from questions that all quorum engines scored. Questions with mixed coverage are excluded from the aggregate but individually reported.

Decay timer: Only advances on COMPLETE or quorum-met runs. PARTIAL and FAILED runs freeze the timer at its current state. This prevents false escalations.

2.4 Database Changes

```
-- Extend ai_audit_runs table
ALTER TABLE ai_audit_runs ADD COLUMN engines_responded SMALLINT DEFAULT 0;
ALTER TABLE ai_audit_runs ADD COLUMN engines_failed JSON DEFAULT '[]';
-- e.g. ['gemini'] or ['gemini','google_sge']
ALTER TABLE ai_audit_runs ADD COLUMN quorum_met BOOLEAN DEFAULT false;
ALTER TABLE ai_audit_runs ADD COLUMN questions_scored SMALLINT DEFAULT 0;
ALTER TABLE ai_audit_runs ADD COLUMN decay_action
  ENUM('advanced','paused','frozen') DEFAULT 'frozen';

-- Extend ai_audit_results for null scores
ALTER TABLE ai_audit_results ALTER COLUMN score DROP NOT NULL;
-- NULL score = question skipped due to engine failure
ALTER TABLE ai_audit_results ADD COLUMN skip_reason VARCHAR(50);
-- Values: null (scored normally), 'timeout', 'rate_limited', 'api_error'
```

Patch 3: Decomposed AI Readiness Scoring

PROBLEM: The 'AI Assistants' channel readiness score is a single 0-100 number with no breakdown. Editors and managers can't tell WHY a score is low or WHAT to fix. Decomposing it into intent-coverage components makes the score actionable.

3.1 AI-Assistant Readiness Components

Component	Weight	Max Score	What It Measures
Answer Block Quality	25%	25	250-300 chars, contains primary intent keyword, is direct answer not marketing, no brand-name lead
FAQ Coverage	20%	20	Number of mandatory FAQ questions answered (from product class template). See Patch 4.
Safety / Regulatory Depth	15%	15	Expert authority block populated + references specific standard (BS 7671, Part P, CE marking)
Cross-SKU Comparison Data	15%	15	Best-For populated (min 2) + Not-For populated (min 1) + Comparison Anchor in AI mapping
Structured Data (JSON-LD)	15%	15	Product schema present + FAQPage schema + Material sameAs (Wikidata) if hero
Citation Audit Score	10%	10	Latest weekly audit: avg score across engines for questions targeting this SKU's cluster

3.2 Computation Logic

```
def compute_ai_readiness(sku):
    components = {}

    # 1. Answer Block Quality (0-25)
    ab = sku.answer_block
    ab_score = 0
    if ab and 250 <= len(ab) <= 300:
        ab_score += 10
        if contains_intent_keyword(ab, sku.primary_intent): ab_score += 8
        if not ab.startswith(BRAND_NAME): ab_score += 4
        if is_direct_answer(ab): ab_score += 3
    components['answer_block'] = ab_score

    # 2. FAQ Coverage (0-20)
    template = get_faq_template(sku.product_class)
    answered = count_answered_faqs(sku.sku_id, template)
    components['faq_coverage'] = round(20 * (answered / len(template)))

    # 3. Safety/Regulatory Depth (0-15)
    safety_score = 0
    if sku.expert_authority: safety_score += 8
    if contains_standard_ref(sku.expert_authority): safety_score += 5
    if sku.compliance: safety_score += 2
    components['safety_depth'] = min(safety_score, 15)

    # 4. Cross-SKU Comparison (0-15)
    comp_score = 0
    if len(sku.best_for) >= 2: comp_score += 5
```

```
    if len(sku.not_for) >= 1: comp_score += 5
    if has_comparison_anchor(sku.cluster_id): comp_score += 5
    components['comparison_data'] = comp_score

    # 5. Structured Data (0-15)
    schema_score = 0
    if has_product_schema(sku.sku_id): schema_score += 6
    if has_faq_schema(sku.sku_id): schema_score += 5
    if sku.wikidata_uri and sku.tier == 'hero': schema_score += 4
    components['structured_data'] = schema_score

    # 6. Citation Audit Score (0-10)
    audit_avg = get_latest_audit_avg(sku.cluster_id)  # 0.0 to 3.0
    components['citation_score'] = round(10 * (audit_avg / 3.0))

    total = sum(components.values())
    return total, components
```

3.3 Database Change

```
-- channel_readiness.component_scores JSON now has defined keys for ai_assistants:
-- {
--   'answer_block': 22,
--   'faq_coverage': 16,
--   'safety_depth': 12,
--   'comparison_data': 15,
--   'structured_data': 11,
--   'citation_score': 7
-- }

-- Dashboard renders these as a radar chart per SKU
```

PASS BARS (unchanged from v2.3.1 but now decomposed)
Hero SKUs: Total ≥ 85. No single component below 50% of its max.
Support SKUs: Total ≥ 70. Answer Block must score ≥ 15/25.
Harvest/Kill: Not scored for AI Assistants channel (readiness = 0).

Patch 4: Mandatory FAQ Templates per Product Class

PROBLEM: We have golden queries at category level for the audit, but no standardised FAQ set that every SKU must answer. LLMs love FAQ pairs — they are the most directly citable content format. Making them mandatory (for Hero/Support) and standardised (per product class) multiplies citability with minimal effort.

4.1 FAQ Templates by Product Class

Cables — Decorative Lighting Cable

#	Question (Standard Wording)	Intent Type	Required Tier	Answer Must Include
1	What cable do I need for [product use]?	Problem-Solving	Hero + Support	Core count + earth wire status + use-case match
2	Is this cable suitable for outdoor use?	Compatibility	Hero + Support	Clear YES/NO + IP rating or Not-For exclusion
3	Do I need 2-core or 3-core cable?	Comparison	Hero + Support	Decision rule: fixed=3-core, portable=2-core. Reference Part P.
4	What length cable do I need?	Specification	Hero + Support	Measure + 30cm rule. Available lengths referenced.
5	Can I install this cable myself?	Installation	Hero only	Skill level statement + Part P reference if fixed. DIY vs professional.
6	What safety certifications does this cable have?	Regulatory	Hero only	CE / UKCA / BS 6500. Reference Expert Authority block.

Cables — Low-Voltage LED Cable

#	Question	Intent	Required Tier	Answer Must Include
1	Can I use this cable for mains wiring?	Compatibility	Hero + Support	Clear NO + max voltage rating + what to use instead
2	What LED driver do I need with this cable?	Specification	Hero + Support	Voltage range + amperage limit
3	What gauge wire for LED strip lights?	Specification	Hero + Support	mm² conductor size + rated amperage at voltage
4	Is this cable safe for indoor LED use?	Regulatory	Hero only	CE marking reference + LED driver requirement

Lampshades (Example)

#	Question	Intent	Required Tier	Answer Must Include

1	Will this shade fit my lamp?	Compatibility	Hero + Support	Fitting type (E27/E14/clip-on) + diameter compatibility
2	Can I use this shade in a bathroom?	Compatibility	Hero + Support	IP rating or clear NO with recommended alternative
3	What wattage bulb can I use?	Specification	Hero + Support	Max wattage + LED equivalent + heat warning if relevant
4	How do I clean this shade?	Problem-Solving	Hero only	Material-specific cleaning instructions
5	What style room does this suit?	Inspiration	Hero only	Room recommendations + interior style references

4.2 Database Schema

```
-- FAQ templates (locked per product class, managed by SEO Governor)
CREATE TABLE faq_templates (
  template_id   VARCHAR(30) PRIMARY KEY,  -- e.g. FAQ-CABLE-DEC-Q1
  product_class VARCHAR(50) NOT NULL,
  question_text VARCHAR(300) NOT NULL,
  intent_type   SMALLINT NOT NULL REFERENCES intent_taxonomy(intent_id),
  required_tier ENUM('hero','support') NOT NULL,  -- Min tier that must answer
  answer_must_include VARCHAR(500) NOT NULL,  -- Guidance for content editor
  ordinal       SMALLINT NOT NULL,
  is_active     BOOLEAN NOT NULL DEFAULT true
);

-- Per-SKU FAQ responses (content team fills these)
CREATE TABLE sku_faq_responses (
  sku_id        VARCHAR(50) NOT NULL REFERENCES sku_master(sku_id),
  template_id   VARCHAR(30) NOT NULL REFERENCES faq_templates(template_id),
  answer_text   VARCHAR(500) NOT NULL,
  updated_at    TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by    VARCHAR(100) NOT NULL,
  PRIMARY KEY (sku_id, template_id)
);

-- Index for readiness computation
CREATE INDEX idx_faq_responses_sku ON sku_faq_responses(sku_id);
```

4.3 Enforcement Integration

Not a publish gate (yet): FAQ completeness is a readiness component (Patch 3), not a hard gate in v2.3.2. Making it a gate in v2.4 once templates are battle-tested.

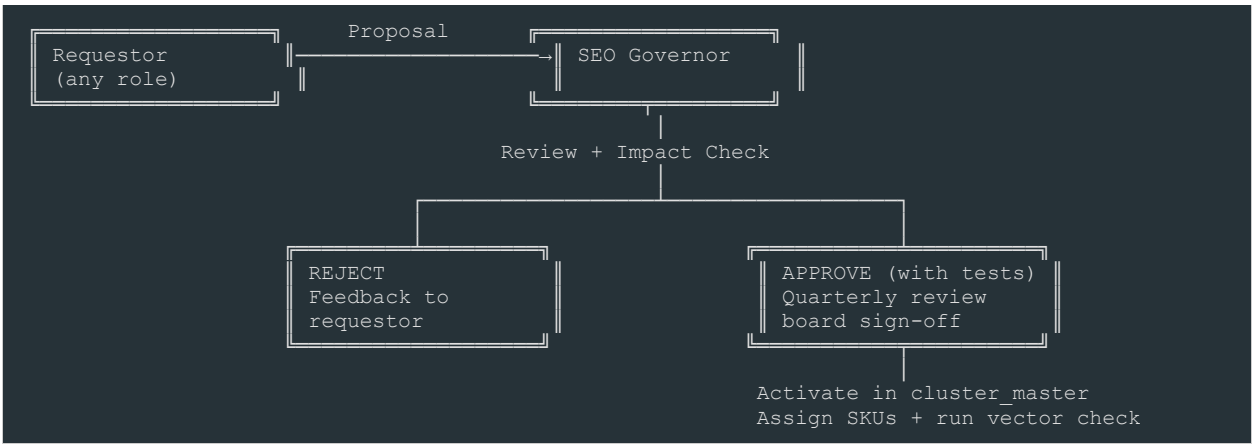
JSON-LD output: For Hero SKUs, all answered FAQs generate FAQPage schema. This is the single highest-impact citability signal for LLMs.

CMS UI: FAQ tab appears on SKU edit page. Questions are pre-populated from the template. Editor writes the answer. 'Answer Must Include' shown as inline guidance.

Patch 5: Cluster Governance Playbook

PROBLEM: The SEO Governor owns the cluster master list, but there is no formal workflow for adding, modifying, or retiring clusters. Without this, cluster bloat will creep in as categories grow. 120+ staff means someone WILL ask for a custom cluster.

5.1 Cluster Change Request Workflow



5.2 Cluster Change Request Requirements

Field Required	What Requestor Must Provide
Proposed Cluster ID	Format: CLU-{CATEGORY}-{TYPE}-{SUBTYPE}. Must not duplicate existing.
Intent Statement	Plain English purpose. e.g. 'Connect LED strip to driver safely for DIY cabinet lighting'
Query Evidence	Minimum 5 real search queries (from Search Console, AI audit, or customer questions) that no existing cluster serves.
SKU Assignment List	Which existing SKUs would move to this cluster, and which would OWN vs SUPPORT.
Impact Assessment	Does this split an existing cluster? If yes, what happens to the parent cluster's SKUs, golden queries, and audit scores?
Commercial Justification	Revenue estimate or margin protection argument. Must reference at least one Hero or Support SKU.

5.3 Governance Rules

Quarterly cadence: Cluster changes reviewed in quarterly taxonomy board (SEO Governor + Commercial Director + Portfolio Holders). No mid-quarter additions unless urgent (commercial impact >£10k/month).

Maximum clusters: No more than 15 clusters per category. If limit reached, must retire/merge before adding.

Retirement: Cluster retirement = soft-delete (is_active=false). All assigned SKUs must be reassigned before deactivation. Golden queries updated in next audit cycle.

Audit trail: Every cluster create/update/retire creates an audit_log entry with the full change request as new_value JSON.

Anti-bloat rule: If a cluster has <3 SKUs after 90 days, it triggers an auto-review flag. Governor must justify retention or merge.

Patch 6: Tier-Mode UX Copy & Banners

PROBLEM: Content editors will encounter hidden/disabled fields for Harvest and Kill SKUs with no explanation. Without clear messaging, they will submit support tickets, attempt workarounds, or assume the system is broken. This spec defines the exact CMS banner text and field-level tooltips per tier.

6.1 CMS Banner Text per Tier

Tier	Banner Colour	Banner Text (Exact Copy)
hero	Green	**HERO SKU — Full CIE Coverage.** This product is a top-revenue performer. All 9 intent types, full Answer Block, FAQ, JSON-LD, and channel feeds are enabled. Target: ≥85 readiness on all active channels within 30 days.
support	Amber	**SUPPORT SKU — Focused Coverage.** This product supports revenue but does not lead. Primary intent + max 2 secondary intents enabled. Answer Block and Best-For/Not-For required. Max 2 hours per quarter.
harvest	Orange	**HARVEST SKU — Maintenance Mode.** This product has low margin and limited growth potential. Only Specification + 1 optional intent are available. Answer Block, Best-For/Not-For, and Expert Authority are suspended. Max 30 minutes per quarter. Focus your time on Hero SKUs instead.
kill	Red	**KILL SKU — Editing Disabled.** This product has negative margin or is flagged for delisting. All content fields are read-only. No time investment permitted. If you believe this classification is wrong, contact your Portfolio Holder to request a tier review (requires Finance co-approval).

6.2 Field-Level Tooltips

Field	Hidden For	Tooltip When Hidden
Secondary Intents 3-9	Harvest, Kill	This field is hidden because your SKU tier only allows Specification + 1 optional intent.
Answer Block	Harvest, Kill	Answer Blocks are suspended for Harvest/Kill tier SKUs. These are reserved for Hero and Support products.
Best-For / Not-For	Harvest, Kill	Comparison data is suspended for this tier. Focus on Specification data only.
Expert Authority	Harvest, Kill	Expert authority signals are reserved for Hero and Support SKUs.
Wikidata URI	Support, Harvest, Kill	Wikidata linking is a Hero-tier feature for maximum AI entity recognition.
FAQ Tab	Harvest, Kill	FAQ content is not required for Harvest/Kill tier SKUs.
All fields	Kill only	This SKU is in Kill tier. All editing is disabled. Contact your Portfolio Holder if you need a tier review.

6.3 Implementation

```
// PHP CMS — render tier banner and field visibility
function render_tier_banner($sku) {
    $banners = [
        'hero'    => ['bg' => '#E8F5E9', 'text' => 'HERO SKU — Full CIE Coverage...'],
        'support' => ['bg' => '#FFF8E1', 'text' => 'SUPPORT SKU — Focused Coverage...'],
        'harvest' => ['bg' => '#FFF3E0', 'text' => 'HARVEST SKU — Maintenance Mode...'],
        'kill'    => ['bg' => '#FFEBEE', 'text' => 'KILL SKU — Editing Disabled...'],
    ];
    $banner = $banners[$sku->tier];
    echo '<div class="cie-tier-banner" style="background:{$banner["bg"]}">';
    echo   '<strong>' . strtoupper($sku->tier) . '</strong> ' . $banner['text'];
    echo '</div>';
}

// Field visibility is ALREADY handled by TIER_FIELD_MAP from v2.3.1 Section 11.2
// This patch adds the tooltip text to the hidden-field div:
function render_hidden_field($field_name, $tier) {
    $tooltips = TIER_TOOLTIPS[$field_name][$tier] ?? null;
    if ($tooltips) {
        echo '<div class="cie-hidden-field-info">' . $tooltips . '</div>';
    }
}
```

Integration Checklist

How these 6 patches integrate with the existing v2.3.1 stack:

#	Patch	DB Changes	API Changes	Sprint Day
1	Fail-Soft Vector	sku_gate_status enum + vector_retry_queue table	ValidationResponse: pending + degraded_mode	Day 1-3 (Phase 1)
2	Audit Degradation	ai_audit_runs + ai_audit_results ALTERs	AuditResults: quorum + degradation fields	Day 8-10 (Phase 2)
3	AI Readiness Decomp	channel_readiness.component_scores schema	ReadinessResponse: component breakdown	Day 11-14 (Phase 2)
4	FAQ Templates	faq_templates + sku_faq_responses tables	New: GET /faq/templates, POST /sku/{id}/faq	Day 15-18 (Phase 3)
5	Cluster Governance	No schema change — operational SOP	No API change — process document	Day 1 (parallel)
6	UX Banners	No schema change — frontend text	No API change — CMS frontend	Day 3-5 (Phase 1)

Updated KPIs (v2.3.2 additions in bold)

#	KPI	Target	Source
1	Publish gate bypass rate	**0%**	Unchanged from v2.3.1
2	AI citation rate (Hero categories)	**>70%**	Unchanged from v2.3.1
3	Kill SKU effort hours	**0 hours**	Unchanged from v2.3.1
4	Vector validation uptime (SAVES not blocked)	**>99.5%**	NEW: Patch 1. Retry queue resolution rate.
5	Vector pending → resolved within 30 minutes	**>95%**	NEW: Patch 1. Retry cron effectiveness.
6	AI audit quorum rate (3+ engines respond)	**>90%**	NEW: Patch 2. Engine availability tracking.
7	Hero SKU FAQ template completion rate	**>80%**	NEW: Patch 4. sku_faq_responses / faq_templates per Hero.
8	Cluster bloat rate (clusters with <3 SKUs after 90d)	**<5%**	NEW: Patch 5. Quarterly governance review.
9	CMS support tickets re: hidden fields / tier confusion	**<5/month**	NEW: Patch 6. UX banner effectiveness.

End of Hardening Addendum.

This addendum patches the 6 gaps identified by cross-engine review. Combined with the v2.3.1 Enforcement Dev Spec, Developer Build Pack, and OpenAPI YAML, your confidence level moves from 93-95% to 97-98%. The remaining 2-3% (multi-language, brand knowledge graph, synthetic query discovery) is deferred to v3.0 as it requires a running system to inform the design.